

PHARMACY SALES DATA ANALYSIS USING PYTHON

A Data Analysis Project

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Date

June 2026

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Introduction

Data analysis plays a vital role in helping businesses make informed decisions by transforming raw data into meaningful insights. In the pharmacy industry, analyzing sales and inventory data enables managers to understand customer purchasing patterns, monitor stock levels, evaluate product performance, and improve overall business operations.

This project focuses on the analysis of a pharmacy sales dataset using Python. The dataset contains information about products sold, product categories, quantities sold, unit prices, total sales, suppliers, stock levels, and product expiry dates. Through data analysis and visualization, valuable insights were generated to identify high-performing products, revenue-generating categories, inventory status, supplier contributions, and daily sales trends.

The analysis was carried out using Python and its data analysis libraries, including Pandas for data manipulation and Matplotlib for data visualization. The findings from this project provide useful information that can support better inventory management, purchasing decisions, sales monitoring, and overall operational efficiency within a pharmacy.

Project Objectives

The main objective of this project is to analyze pharmacy sales data using Python in order to generate meaningful business insights that support effective decision-making.

The specific objectives are to:

1. Determine the total, average, highest, and lowest revenue generated from pharmacy sales.
2. Identify the highest- and lowest-performing products based on total revenue.
3. Analyze product categories to determine their revenue contribution and quantity sold.
4. Evaluate supplier performance by examining their contribution to total sales revenue.
5. Assess inventory levels by identifying products with high stock, low stock, and those requiring restocking.
6. Monitor product expiry dates by identifying expired products and products that will expire within the next 30 days.
7. Analyze daily sales trends to identify periods of high and low sales performance.
8. Identify fast-moving and slow-moving products to support inventory planning.
9. Create charts and visualizations that clearly communicate the findings of the analysis.
10. Provide recommendations that can help improve inventory management, sales performance, and overall pharmacy operations.

Tools Used

The following tools and technologies were used to complete this project:

- * Python – Used as the primary programming language for data analysis.
- * Pandas – Used for data loading, cleaning, manipulation, and analysis.
- * Matplotlib – Used to create charts and visualizations.
- * Microsoft Excel – Used to store and inspect the pharmacy sales dataset.
- * Google Docs – Used to prepare the project report.
- * GitHub – Used to host and share the completed project.

Dataset Description

The dataset used in this project is a pharmacy sales dataset that contains records of medicines sold in a pharmacy. The data was cleaned and prepared before analysis to ensure accurate results.

The dataset includes information about sales transactions, product details, pricing, stock availability, suppliers, and expiry dates. It was analyzed using Python to identify sales trends and generate business insights that can support inventory management and decision-making.

The main columns in the dataset include:

- Date
- Invoice Number
- Product Name
- Category
- Quantity Sold
- Unit Price
- Total Sales
- Supplier
- Expiry Date
- Stock in Hand.

Methodology

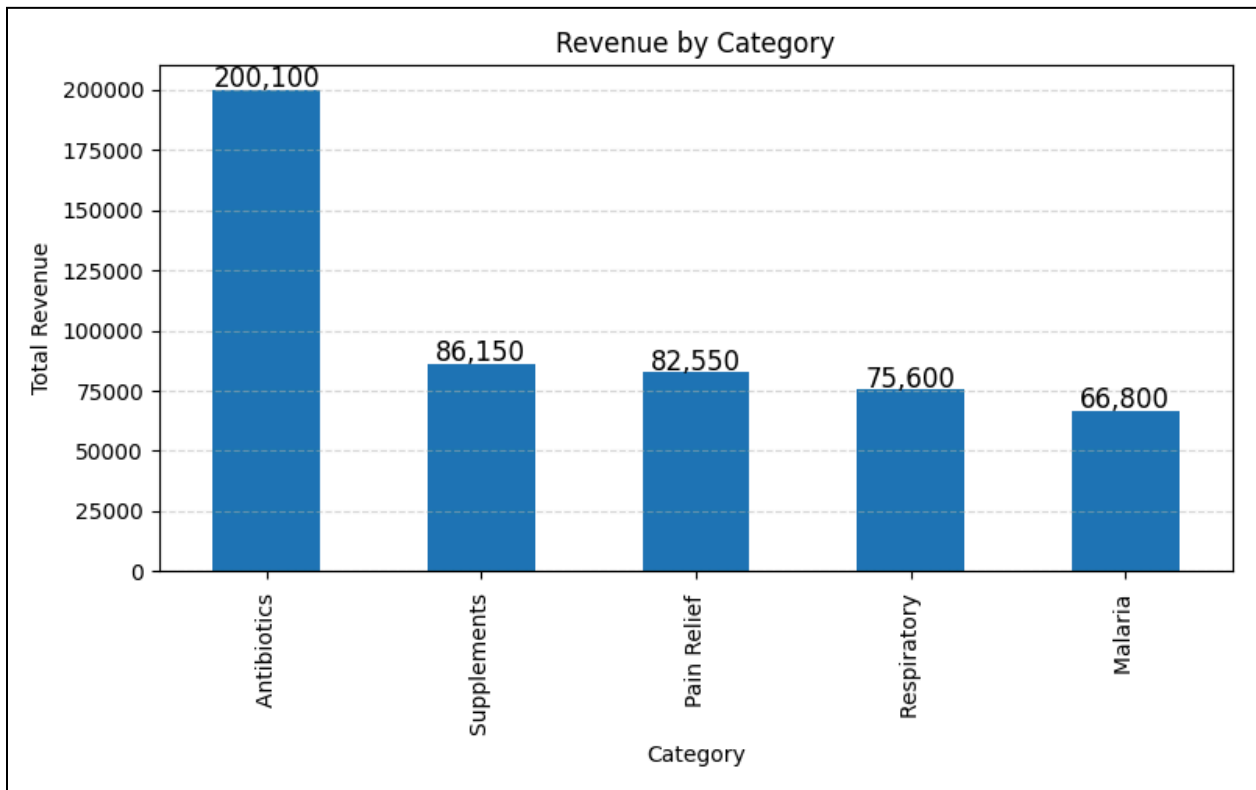
The pharmacy sales dataset was analyzed using Python. The analysis followed a structured process to ensure accurate and meaningful results.

The steps involved were:

1. The dataset was imported into Python using the Pandas library.
2. The data was inspected to check its structure, data types, and missing values.
3. Data cleaning was performed to ensure the dataset was ready for analysis.
4. Descriptive statistics were generated to summarize the data.
5. Sales performance, product categories, stock levels, and other business metrics were analyzed.
6. Charts and visualizations were created using Matplotlib to present the findings in a clear and understandable manner.
7. The results were interpreted to provide insights and recommendations for improving pharmacy sales and inventory management.

Data Analysis and Results

Figure 1: Revenue by Category

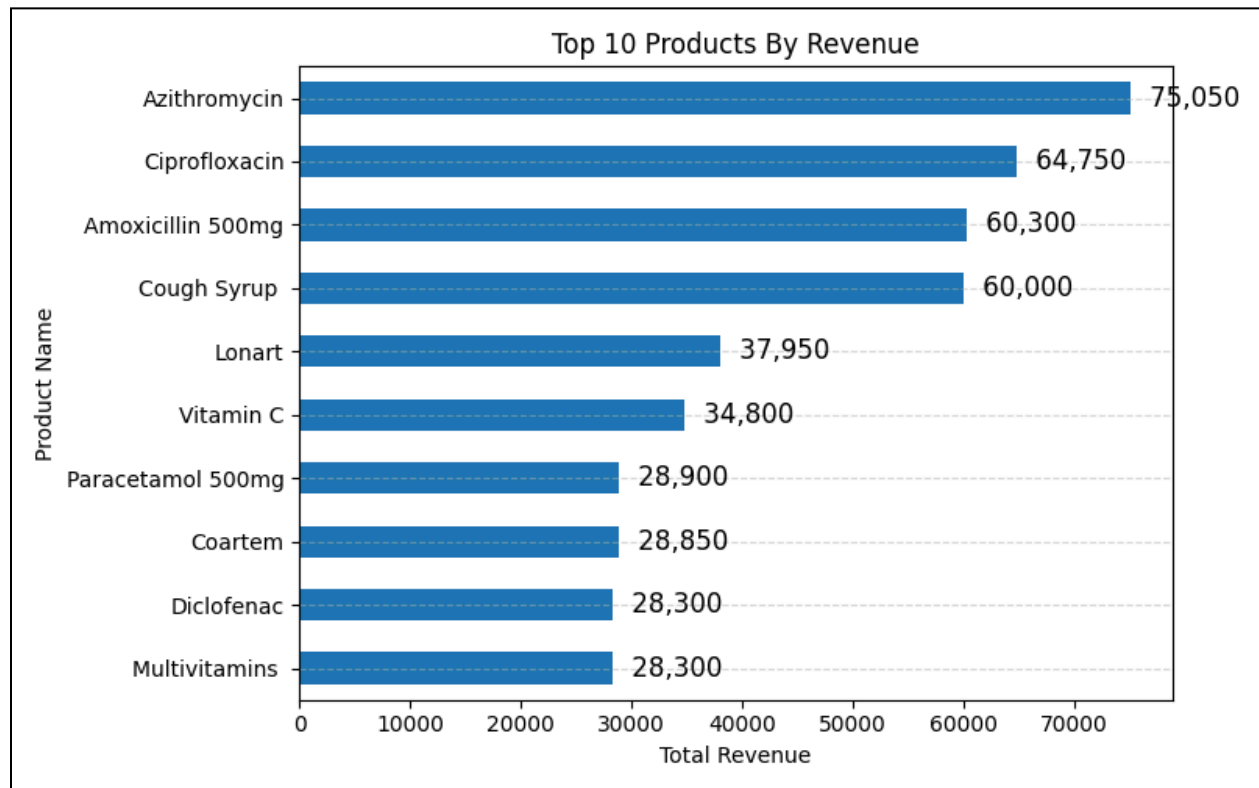


This chart illustrates the total revenue generated by each product category in the pharmacy. The analysis groups all sales by category and calculates the total revenue contributed by each one.

The chart helps identify the categories that generate the highest income for the pharmacy. Categories with higher revenue contribute significantly to overall business performance, while lower-performing categories may require promotional strategies or inventory review.

This analysis enables pharmacy managers to understand which categories drive sales and supports better decision-making regarding stock allocation and product marketing.

Figure 2: Top 10 Products by Revenue

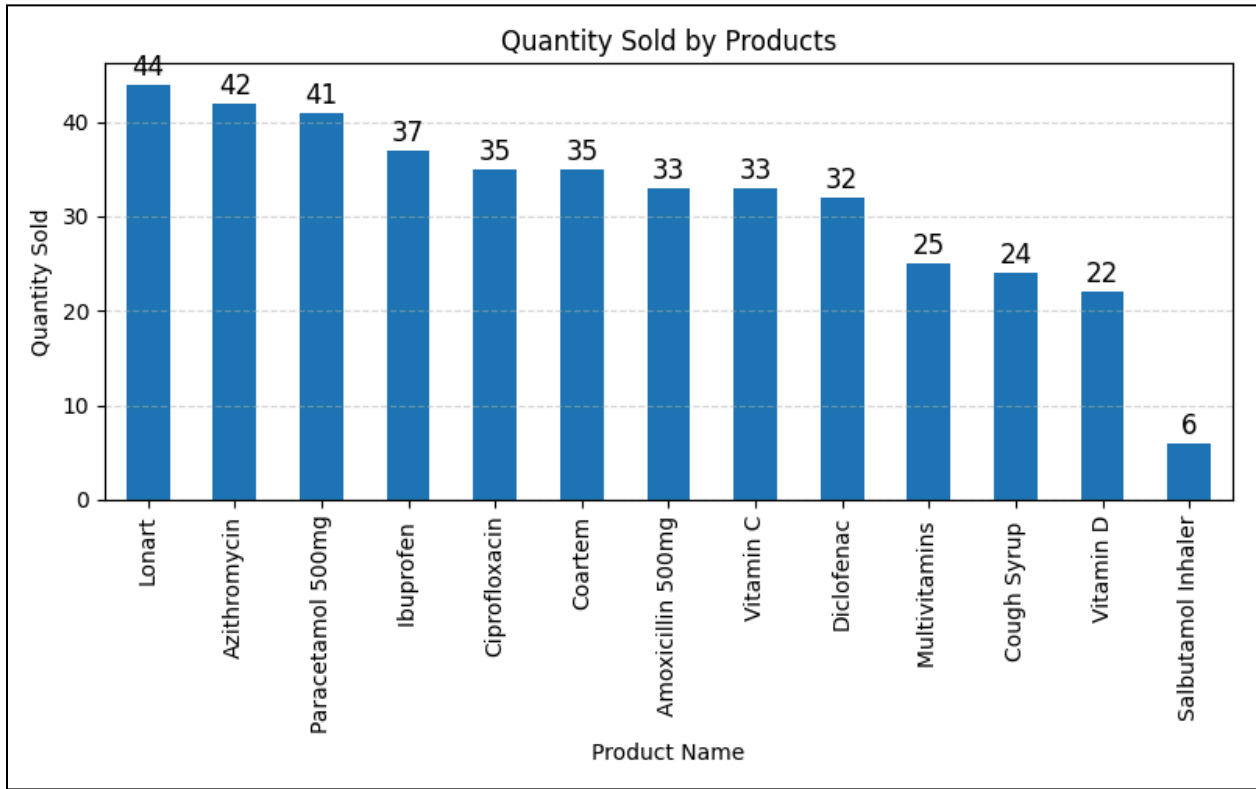


This chart presents the top ten products that generated the highest total revenue during the analysis period. The products are ranked from the highest to the lowest based on the total sales value.

The chart helps identify the pharmacy's highest-earning products and highlights the medicines that contribute most to overall revenue. These products are important for maintaining profitability and should be monitored to ensure adequate stock levels.

The results can help pharmacy managers prioritize inventory planning, purchasing decisions, and promotional activities for high-performing products while also evaluating the performance of lower-selling products.

Figure 3: Quantity Sold by Products

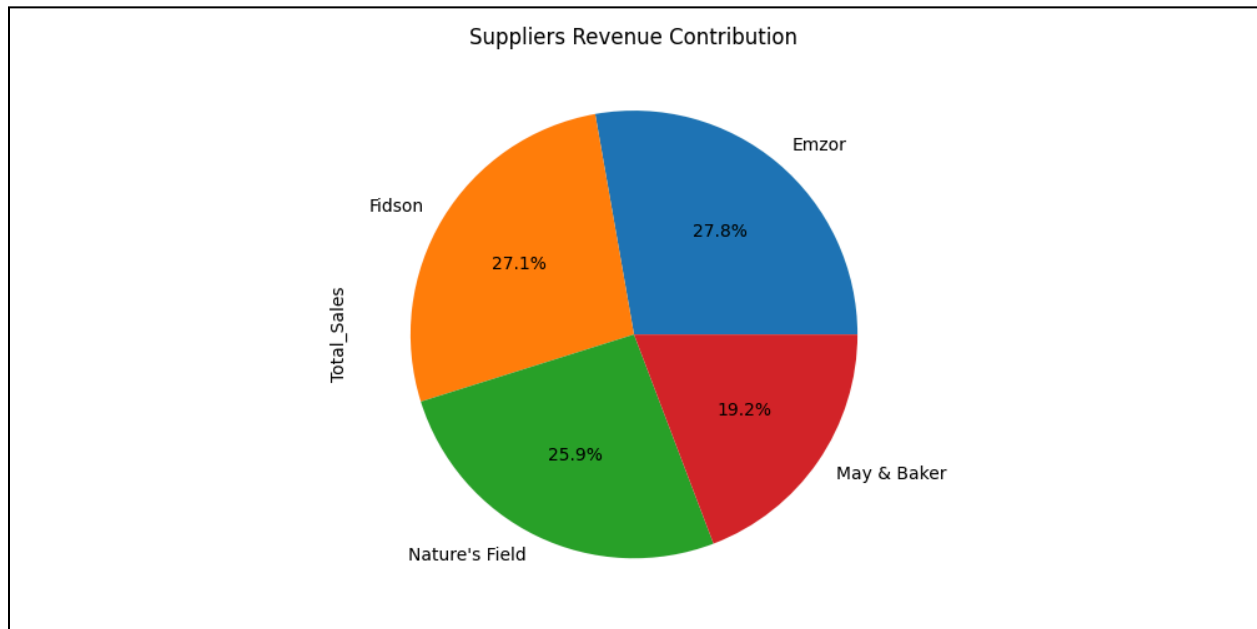


This chart displays the total quantity sold for each product in the pharmacy during the analysis period. It compares the sales volume of different products based on the number of units purchased by customers.

The chart helps identify the most frequently purchased products, which are considered fast-selling items. Products with high sales volumes require regular stock monitoring to prevent shortages, while products with lower sales volumes may require promotional strategies or a review of purchasing patterns.

The findings from this analysis support effective inventory management by helping the pharmacy maintain adequate stock levels for high-demand products and reduce the risk of overstocking slow-moving items.

Figure 4: Supplier Revenue Contribution

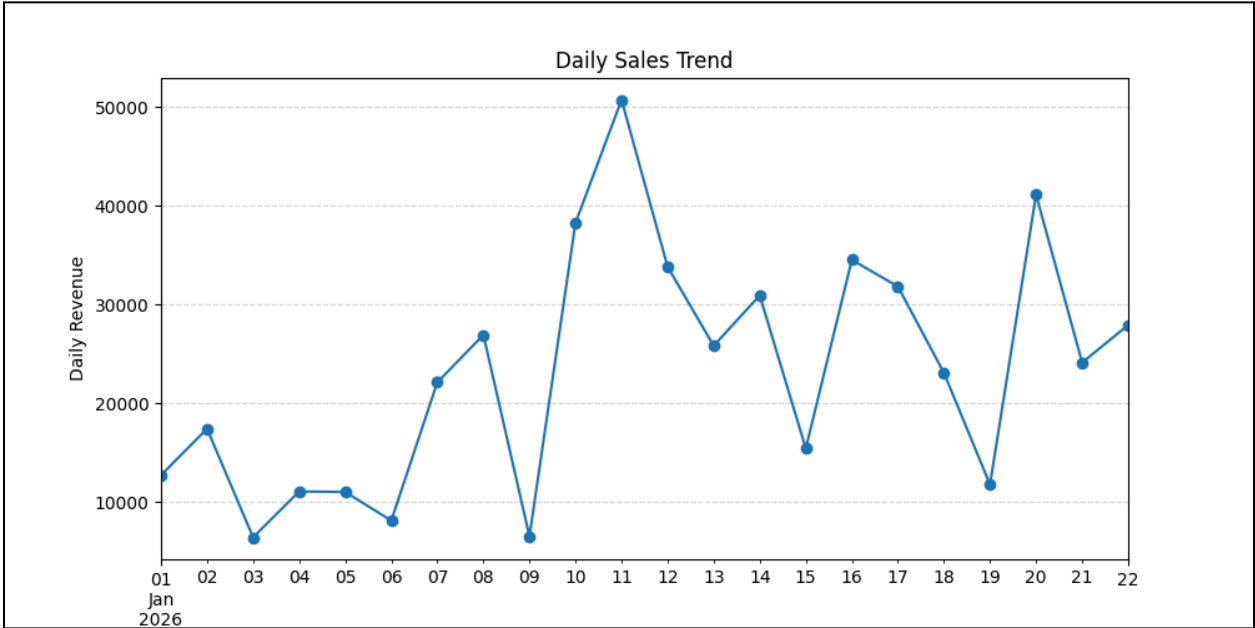


This pie chart illustrates the percentage contribution of each supplier to the pharmacy's total revenue. It shows how much revenue is generated from products supplied by each supplier, making it easier to compare their overall business contribution.

The analysis helps identify the suppliers that contribute the highest share of revenue. Suppliers with larger contributions play a significant role in the pharmacy's sales performance and should be considered strategic partners. Conversely, suppliers with smaller contributions may require further evaluation to determine whether their products should be promoted more effectively or reviewed for profitability.

This information supports supplier performance evaluation, purchasing decisions, and the development of stronger supplier relationships to improve business performance.

Figure 5: Daily Sales Trend

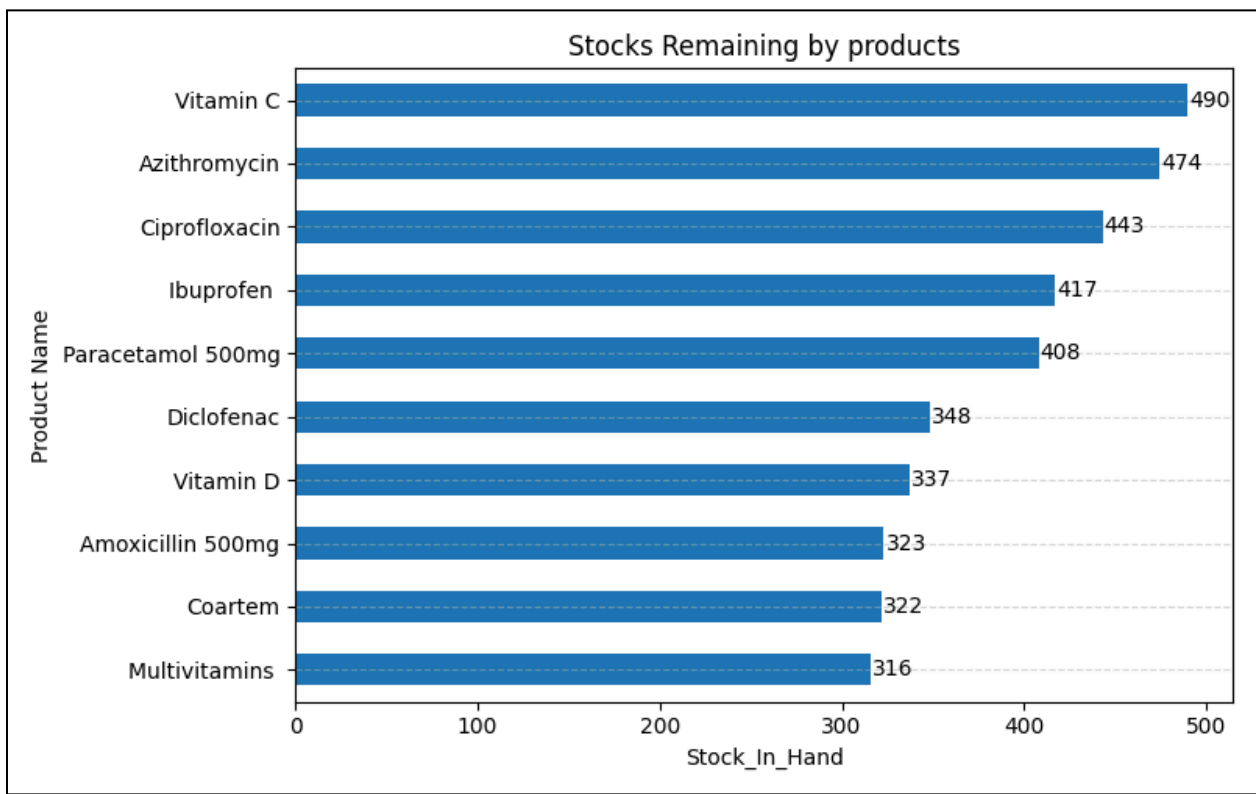


This line chart illustrates the daily sales trend of the pharmacy over the analysis period. It shows how total revenue changed from one day to another, making it easier to identify fluctuations in daily sales performance.

The chart highlights days with higher and lower sales, which can help the pharmacy understand customer purchasing patterns and identify peak sales periods. These insights are useful for planning inventory levels, staffing requirements, and promotional activities during periods of high demand.

Monitoring daily sales trends also enables the pharmacy to evaluate business performance over time and make informed decisions aimed at improving revenue and operational efficiency.

Figure 6: Stocks Remaining by Products

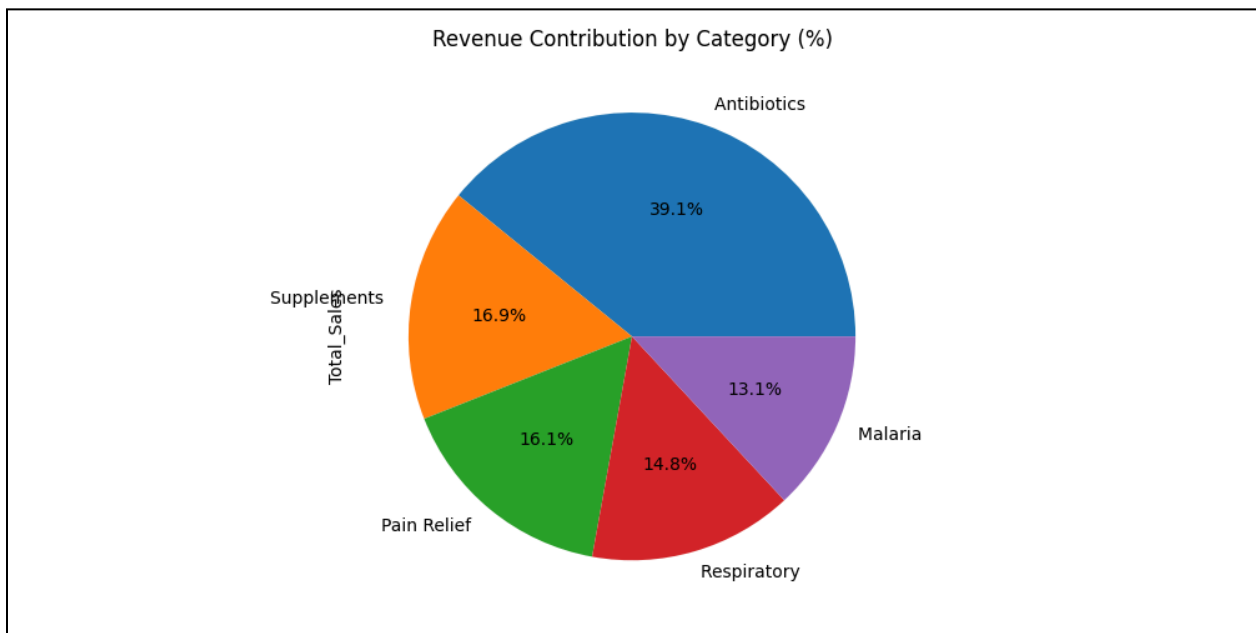


This chart presents the remaining stock available for the top products in the pharmacy. It compares the quantity of each product currently in stock, providing a clear view of inventory levels.

The analysis helps identify products with high stock levels and those that may require restocking soon. Maintaining an appropriate inventory level is essential to ensure product availability, avoid stock shortages, and minimize the risk of overstocking.

The results support effective inventory management by enabling pharmacy managers to monitor stock levels, plan future purchases, and ensure that essential medicines remain available to meet customer demand.

Figure 7: Revenue Contribution by Category (%)

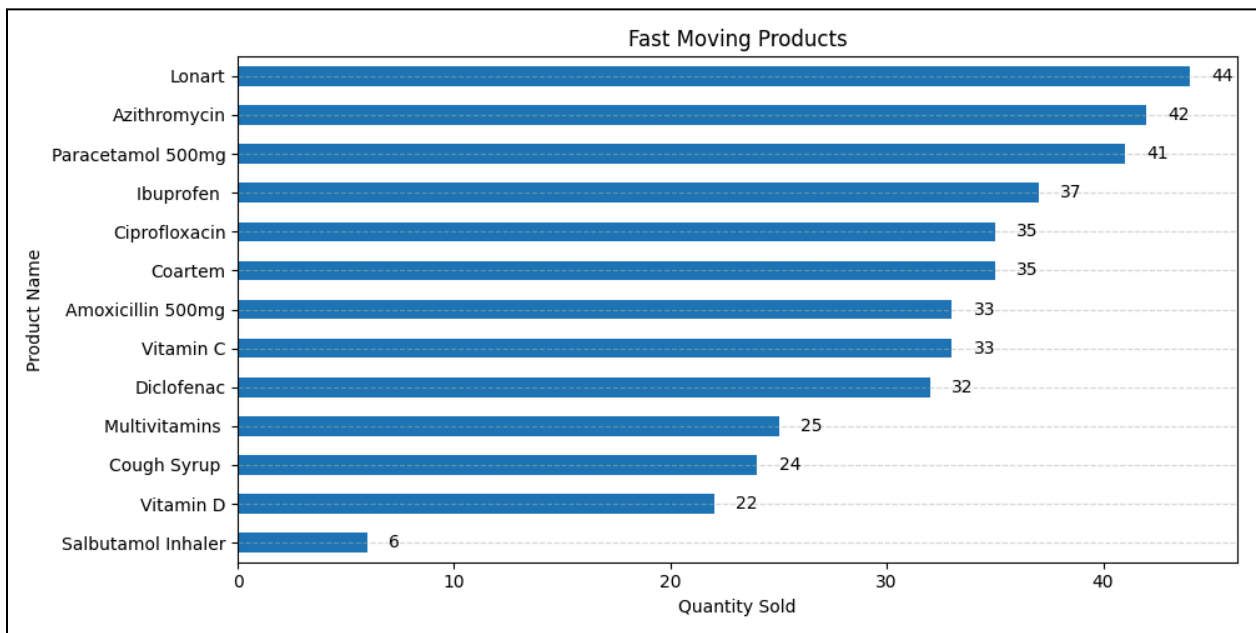


This pie chart illustrates the percentage contribution of each product category to the pharmacy's total revenue. It provides a clear comparison of how much each category contributes to overall sales performance.

The analysis identifies the categories that generate the largest share of revenue and those with smaller contributions. Understanding the revenue distribution across categories helps pharmacy managers focus on high-performing product groups while identifying categories that may require additional marketing efforts or inventory adjustments.

These insights support strategic business planning, product management, and revenue optimization by ensuring resources are allocated to the categories that contribute most to the pharmacy's financial performance.

Figure 8: Fast Moving Products



This chart displays the products with the highest quantities sold during the analysis period. It highlights the medicines and pharmacy products that experience the greatest customer demand.

Identifying fast-moving products is essential for effective inventory management. These products should be monitored closely to ensure they remain in stock and are reordered before inventory levels become critically low. Maintaining sufficient stock of high-demand products helps improve customer satisfaction and prevents lost sales.

The findings also provide valuable insights for purchasing decisions, demand forecasting, and inventory planning, enabling the pharmacy to optimize stock levels while reducing the risk of shortages.

Key Findings

The analysis showed that some product categories generated significantly higher revenue than others, indicating their importance to the pharmacy's overall sales performance. A small number of products contributed a large share of total revenue and were also among the fastest-moving items. Supplier analysis revealed that certain suppliers accounted for a greater proportion of sales, making them important business partners. Inventory analysis identified products with both high and low stock levels, highlighting the need for regular stock monitoring and timely restocking. The daily sales trend demonstrated fluctuations in revenue, providing useful information for planning inventory, staffing, and promotional activities.

Conclusion

This project successfully analyzed pharmacy sales data using Python to generate meaningful business insights. By examining revenue, product performance, supplier contribution, inventory levels, and sales trends, the analysis demonstrated how data can support informed decision-making. The visualizations made it easier to identify important patterns that can improve inventory management, sales performance, and overall business operations. The project also demonstrates practical skills in data cleaning, analysis, and data visualization using Python.

Recommendations

Based on the findings of this analysis, the following recommendations are proposed:

1. Maintain adequate stock levels for fast-moving and high-revenue products to prevent stock shortages.
2. Regularly monitor products with low stock levels and restock them before they run out.
3. Strengthen relationships with suppliers that consistently contribute high-value products while periodically reviewing the performance of lower-contributing suppliers.
4. Closely monitor products approaching their expiry dates to reduce inventory losses through timely sales or stock management.
5. Continue tracking daily sales trends to improve demand forecasting, inventory planning, and business decision-making.
6. Perform regular data analysis to monitor business performance and identify opportunities for continuous improvement.

